BMO1 2025 (November 2025), Problem 2

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The problem

Find all triples (x, y, z) of real numbers satisfying the equations

$$x^2 + 2yz = 4, \qquad y^2 + 2zx = 4, \qquad z^2 + 2xy = 1.$$

The answer is $(x, y, z) = (0, 2, 1), (0, -2, -1), (2, 0, 1), (-2, 0, -1)$.

Script A **10/10**

Subtracting the second equation from the first,

$$x^2 - y^2 - 2z(x - y) = 0 \implies (x - y)(x + y - 2z) = 0,$$

so $x = y$ or $x + y = 2z$.

Case 1: *$x = y$. The equations become $x^2 + 2xz = 4$ and $z^2 + 2x^2 = 1$. Note that $x \neq 0$ (else $0 = 4$), so $z = \dfrac{4 - x^2}{2x}$. Substituting and multiplying by $4x^2$,*

$$(4 - x^2)^2 + 8x^4 = 4x^2 \implies 9x^4 - 12x^2 + 16 = 0.$$

As a quadratic in x^2 the discriminant is $144 - 576 < 0$, so there is no real x and no solution in this case.

Case 2: *$x + y = 2z$. Substituting $z = \frac{1}{2}(x + y)$, the first two equations both become $x^2 + xy + y^2 = 4$ and the third becomes $x^2 + 10xy + y^2 = 4$. Hence $9xy = 0$. If $x = 0$ then $y = \pm 2$ and $z = \frac{1}{2}y = \pm 1$. If $y = 0$ then $x = \pm 2$ and $z = \frac{1}{2}x = \pm 1$ (same signs).*

Checking in all three equations, $(0, 2, 1), (0, -2, -1), (2, 0, 1), (-2, 0, -1)$ are solutions, and these are all of them.

Marker's comment. Complete. Both factors are kept, $x \neq 0$ is checked before dividing, and the case $x = y$ is closed by the discriminant, which most scripts skip. The final check confirms the four triples.

Script B **7/10**

Subtracting the first two equations gives $(x - y)(x + y - 2z) = 0$.

If $x = y$ then $x^2 + 2xz = 4$ and $z^2 + 2x^2 = 1$. Here $x \neq 0$, so $z = (4 - x^2)/2x$, and substituting gives $9x^4 - 12x^2 + 16 = 0$. The discriminant in x^2 is $144 - 576 < 0$, so there are no real solutions.

If $x + y = 2z$, substituting $z = \frac{1}{2}(x + y)$ into the equations gives $x^2 + xy + y^2 = 4$ and $x^2 + 10xy + y^2 = 4$, so $xy = 0$ and $x^2 + y^2 = 4$. From the third equation $z^2 = 1 - 2xy = 1$, so $z = \pm 1$. Therefore the solutions are

$$(0, \pm 2, \pm 1), \qquad (\pm 2, 0, \pm 1),$$

eight triples in all.

Marker's comment. The casework and the hard case are right. At the end z is found again from the third equation, which drops the link $z = \frac{1}{2}(x+y)$, so four of the eight triples are false. For example $(0,-2,1)$ gives -4 in the first equation, and substituting the answers back would have caught the slip.

Script C 4/10

Subtracting the second equation from the first gives $(x-y)(x+y-2z)=0$. *Dividing by* $x-y$, *we get* $x+y=2z$.

Then $z=\frac{1}{2}(x+y)$ *and the equations become* $x^2+xy+y^2=4$ *(twice) and* $x^2+10xy+y^2=4$. *Subtracting,* $9xy=0$. *If* $x=0$ *then* $y=\pm 2$ *and* $z=\pm 1$, *and if* $y=0$ *then* $x=\pm 2$ *and* $z=\pm 1$, *with the signs matching since* $z=\frac{1}{2}(x+y)$. *All four of* $(0,2,1)$, $(0,-2,-1)$, $(2,0,1)$, $(-2,0,-1)$ *satisfy the original equations, so these are the solutions.*

Marker's comment. The answer and Case 2 are right, but dividing by $x-y$ discards the case $x=y$ unexamined. The case $x=y$ is the harder half and needs the discriminant argument of Script A. Without this case the script only shows the four triples are solutions, not that there are no others.

Script D 1/10

Look for integer solutions. The third equation $z^2+2xy=1$ *suggests* $z=\pm 1$ *and* $xy=0$. *If* $x=0$ *the first equation gives* $2yz=4$, *so* $yz=2$, *and the second gives* $y^2=4$. *So* $y=2$, $z=1$ *or* $y=-2$, $z=-1$. *Similarly if* $y=0$ *then* $x=\pm 2$ *and* $z=\pm 1$ *with the same sign. These four triples satisfy all three equations, so the solutions are* $(0,2,1)$, $(0,-2,-1)$, $(2,0,1)$, $(-2,0,-1)$.

Marker's comment. The problem asks for real solutions and the script searches only integers. Even that search is incomplete, since $z=3$, $xy=-4$ is never ruled out. The mark is for finding and checking the four solutions.

What full marks needs

Factorise to $(x-y)(x+y-2z)=0$ and keep both cases. The case $x=y$ ends with the negative discriminant of $9x^4-12x^2+16$ as a quadratic in x^2, and the case $x+y=2z$ keeps $z=\frac{1}{2}(x+y)$ to the end, with the answers checked in all three equations.